

Research statement

Overview

I am an applied micro-economist working at the intersection of health economics, energy economics, and public policy. My research is organized around the question: how do information environments, institutional and market structures shape the decisions people and firms make, and what are the consequences of those decisions? I pursue these questions with causal econometric methods: difference-in-differences, event-study designs, instrumental variables, and longitudinal panel analysis applied to large administrative and survey datasets. I pursue two distinct research streams from this agenda. One addresses problems in healthcare delivery, ranging from individual-level outcomes in adolescent health and chronic disease management in adults to the adoption of Artificial Intelligence (AI) in U.S. hospitals. The other stream is on the economics of critical minerals.

Research Stream I: Social Media, Health Policy, and AI Adoption in U.S. Hospitals

My research examines how information environments, public policy, and market institutions shape economic and health outcomes. My work uses advanced econometric methods and nationally representative and administrative data to study energy policy, adolescent health behaviors, and the adoption of artificial intelligence in healthcare settings. This has produced multiple peer-reviewed publications in top journals like *Archives of Physical Medicine and Rehabilitation*.

Other papers are currently under review. These include work on adolescent mental health responses to recreational marijuana legalization. I also have two papers under review examining AI adoption in U.S. psychiatric hospitals and long-term healthcare facilities, work that required me to engage deeply with the organizational structure of behavioral health service delivery, payer mix, and the regulatory environment governing inpatient psychiatric care.

A separate project in this stream examines the dynamic effects of Medicaid expansion on long-run access to care for low-income adults with chronic conditions. Causal identification relies on the staggered adoption of expansion across states and an event-study design that allows the treatment effect to evolve. I view this paper as a bridge between adolescent behavior research and a longer-term interest in how insurance markets and coverage policies shape healthcare access for vulnerable populations.

Research Stream II: The Economics of Critical Minerals and Trade Policy

My second research stream investigates the macroeconomic and strategic implications of global Rare Earth Element (REE) markets: a topic at the center of policy discussions as supply chain vulnerabilities have become a national security concern. I entered this area because the economics literature lacked credible empirical estimates of key quantities that policymakers urgently need: how large are the security costs embedded in REE supply chains, and does trade policy move market power dynamics in this sector?

My paper in this stream provides the first-ever empirical estimates of security premiums: costs not reflected in market prices associated with REE supply shocks for the United States and G10 countries. The estimates are derived by exploiting variation in supply disruptions and measuring the gap between market prices and the full social cost of REE consumption, including the cost of import dependence. The practical implication is that current market prices substantially understate the strategic value of domestic REE development and supply diversification.

Future Research Agenda

Over the next five years, I plan to deepen both streams. In the health and social media stream, my immediate priority is to advance the R&R papers to publication and to extend the marijuana legalization work into a more general framework for studying how state-level policy changes propagate through adolescent risk behavior over time. I am actively seeking collaboration with researchers who have access to claims data or administrative health records, and I believe the Medicaid expansion paper opens doors in that direction.

In the critical minerals stream, I plan to build on the REE security premium estimates by modeling the long-run effects of supply chain diversification strategies on energy transition costs. As battery technology and renewable energy infrastructure scale up, the demand structure for REEs and other critical minerals is shifting in ways that existing models do not capture well.

I intend to actively pursue external funding from NSF, NIH (for health work), and DoE-affiliated programs (for critical minerals work) as I build my independent research program. My goal is to produce empirical work that is methodologically rigorous, broadly accessible, and consequential for the policy conversations it engages.